4 (a) A beam of light from a sodium lamp passes through a pair of narrow slits S_1 and S_2 producing a pattern on a screen as shown in Fig. 4.1. The pattern on the screen consists of regularly spaced bright and dark fringes, as shown in Fig. 4.2.

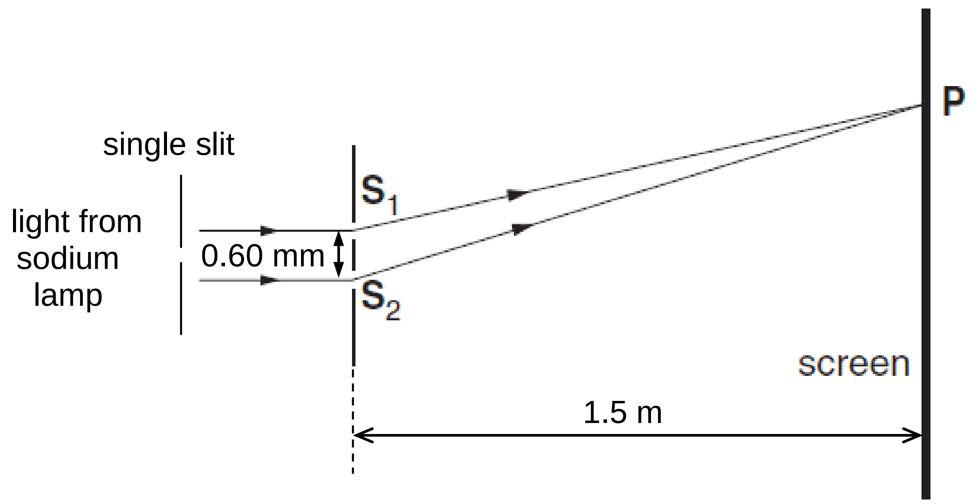

Fig. 4.1 (not to scale)

(i) State and explain the conditions necessary for the light from the two slits S_1 and S_2 to produce a visible pattern on the screen.

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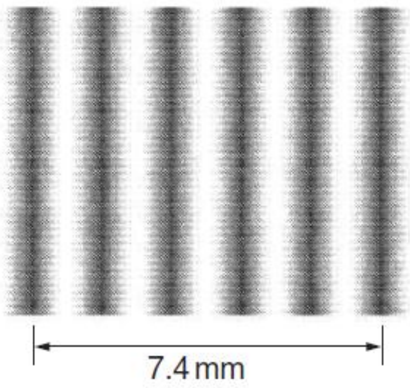
…………………………………………………………………………………… [3]

(i) **Solution:**

Any of the 3:

1. Waves from the two slits, S_1 and S_2, must be coherent so that they have a constant phase difference to produce a steady pattern on the screen. **1**
2. A single slit must be placed between the two slits and the sodium lamp to ensure that S_1 and S_2 acts as coherent sources. **1**
3. Amplitudes or intensities of the two waves must be approximately equal. Otherwise, complete cancellation cannot occur at positions of destructive inference. **1**
4. Since $x = \dfrac{\lambda D}{a}$, distance between the two sources must be small compared to the wavelength of the wave. This is to ensure that the fringe separation is large enough to distinguish the maxima. **1**
5. The slits must be narrow or close together so that the diffraction patterns from the slits can overlap. **1**

(i) Explain the condition for a bright fringe to appear on the screen at P in Fig. 4.1.

			…………………………………………………………………………… …………………………………………………………………………… …………………………………………………………………………… …………………………………………………………………………… …………………………………………………………………………… ……………………………………………………………………………	[3]
		1.	**Solution:** The two waves from the slits, S_1 and S_2, must arrive in phase at the position P, where constructive interference occurs to give the maximum amplitude. The path difference between the two slits and the position of P must be an integer number of wavelengths for the waves to arrive in phase at P.	**1** **1** **1**
	(b)	Fig. 4.2 shows the central part of the fringe pattern on the screen from the slits S_1 and S_2.  **Fig. 4.2** Calculate		
		(i)	the fringe separation x, x = ……….………………… mm	[1]
		(i)	**Solution:** $x = \dfrac{7.4}{5} = 1.48$ mm	**1**
		(ii)	the wavelength λ of the yellow light.	

			λ = ……………..…………………….. m	[2]
		(ii)	**Solution:** Using $x = \dfrac{\lambda D}{a}$ $\Rightarrow \lambda = \dfrac{ax}{D} = \dfrac{\left(0.60\times10^{-3}\right)\left(1.48\times10^{-3}\right)}{1.5}$ $= 5.92 \times 10^{-7}$ m	1 1
	(c)	The light from the sodium lamp is analysed using an instrument containing a diffraction grating. The diffraction grating has 500 lines per millimeter. Calculate the angle of diffraction of the spectral line in the second order spectrum. angle = ……………………………..……^{0}		[2]
	(c)	**Solution:** $d = \dfrac{1}{N} = \dfrac{10^{-3}}{500} = 2\times10^{-6}$ m Using $d\sin\theta = n\lambda$ $\sin\theta = \dfrac{n\lambda}{d} = \dfrac{2\times5.92\times10^{-7}}{2\times10^{-6}} = 0.592$ $\theta = 36.299^{0} = 36.3^{0}$		1 1

5	(a)	State Faraday's law of electromagnetic induction.	